
HICUM/L2 v3.2.0 - Release Notes

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Overview

v3.2.0 beta1 QA package uploaded on 8/25/2025

v3.2.0 beta2 QA package uploaded on 11/12/2025

v3.2.0 beta3 QA package uploaded on 12/17/2025

v3.2.0 QA package uploaded on 6/5/2026

Release Notes Content

- Coding related changes
- Device physics related changes
- Summary of changes
- Changes in qaSpec

Coding related changes

Workgroup list items Q4_2023.1, Q2_2024.4

Declaration of physical constants **based on NIST2018 values**

- **physical constants defined in code**

```
`define P_QEL_2018 1.602176634e-19
`define P_KB_2018 1.380649e-23
:
if (flcomp <= 310) begin
    P_QEL = `P_QEL_310;
    P_KB  = `P_KB_310;
end else begin
    P_QEL = `P_QEL_2018;
    P_KB  = `P_KB_2018;
end
```

- **declaration of thermal node changed to *thermal* instead of *electrical*:**

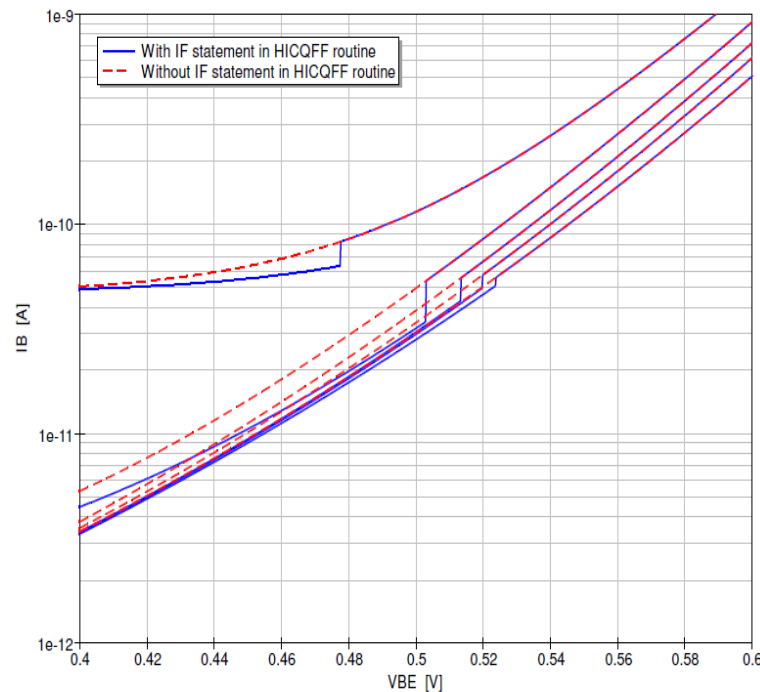
```
thermal    tnode;
```

Also: renamed all related branch voltages and currents (V,I) -> (Temp,Pwr)

```
Pwr(br_sht) <+ Temp(br_sht)/rth_t-pterm;
:
```

Code related changes (Q2_2024.5)

- Discontinuity observed in low injection base current



- BC barrier effect: should be visible only at *very high* injection (here non-physical time constant TBHREC):

```
\define HICQFF \
```

```
:
```

```
if (itf >= 1.0e-6*ick) begin \
```

```
:
```

```
Q_bf      = FFdQbfb+FFdQbfc; \
```

```
:
```

conditional block for reducing simulation time

=> removed conditional statement

- related: need to avoid zero transfer current in previous conditional block

```
FFdTef = tef0_t*exp(gtfe*ln(FFitf_ick)); \
```

by changing lower bound of model parameter c100

```
\MPRoc( c100 , 2.0E-30 , "A^2s" , 0 , 1 , "GICCR constant")
```

- removed several potential divisions by zero by rearranging expressions

Code related changes (cont'd)

GMIN in collector-substrate branch

- **v310**: GMIN is added to collector-substrate junction branch

Requirement:

- CS junction current and GMIN should only be evaluated for *junction isolation* but should be zero for *oxide isolation* (e.g. SOI process)

=> **code change in v320**:

- For **oxide isolation**: model parameter **iscs** = 0 => skip GMIN addition:
- **implementation**:

```

if (iscs > 0.0E0) begin // evaluate only if CS junction exists
    I(br_sici) <+ type*ijsc;
    if (flcomp >= 310) begin
        I(br_sici) <+ gmin*V(br_sici);
    end
end
end

```

Coding related changes (cont'd)

- **relative tolerance for transfer current iteration** made consistent with simulator current tolerance => **changed**

```
`define RTOLC          1.0e-5
to
`define RTOLC          $simparam("reltol",1.0e-5)
```

=> will not cause differences beyond specified QA tolerances

- replaced `Gmin by **real variable "gmin"** at all locations
- **inconsistency between implementation and documentation** of intrinsic output conductance **for OP output** => **changed**

```
g_Oi= ddx(it,V(ci));
to
g_Oi= ddx(it,V(ci))-g_avr;
```

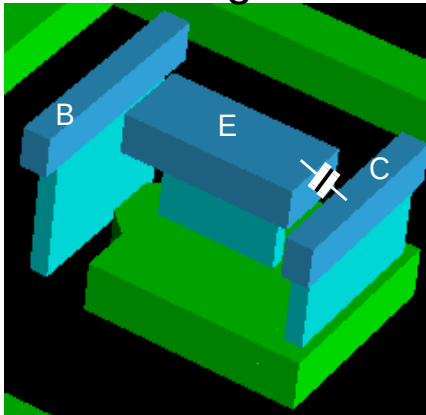
=> **now consistent with documentation**

Device physics related changes

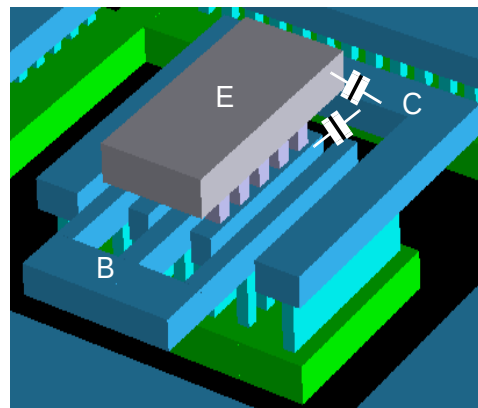
Added parasitic CE capacitance **CCEPAR** (Q4_2024.6)

- representing metal connection related capacitance within basic cell; i.e. metal layers that make transistor functional before being connected in a circuit)
- relevant for output matching in amplifiers (e.g., switch-mode power amplifiers)

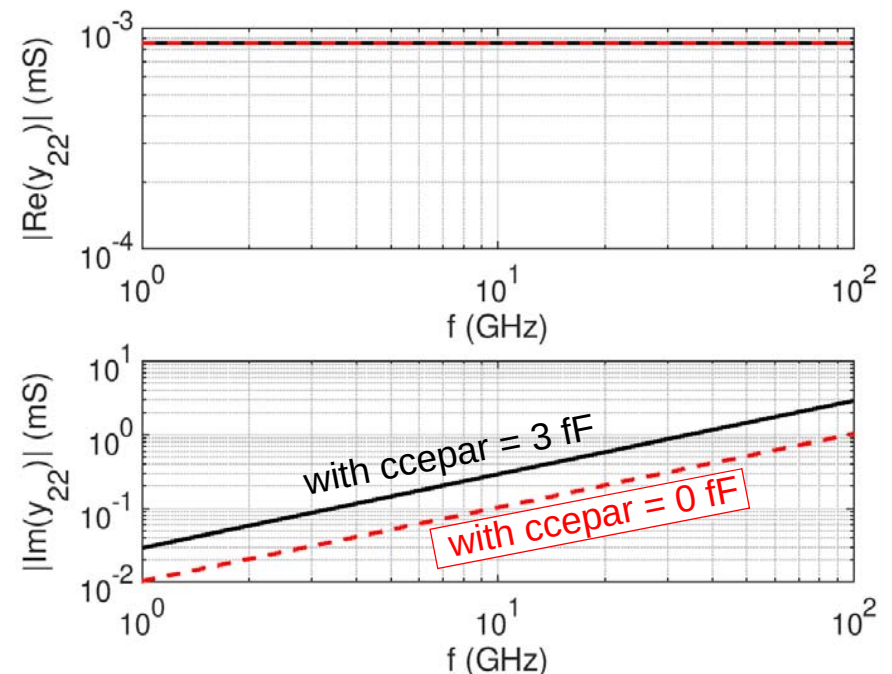
BEC configuration



CBEBEBC config.



Implementation test results
(ccepar = 3 fF)



Additional weight factors (Q2_2024.3)

Purpose:

- compensation of incorrect values of Qp0
- decoupling of model parameter extraction for DC and small-signal curves
- improving accuracy in reverse operation

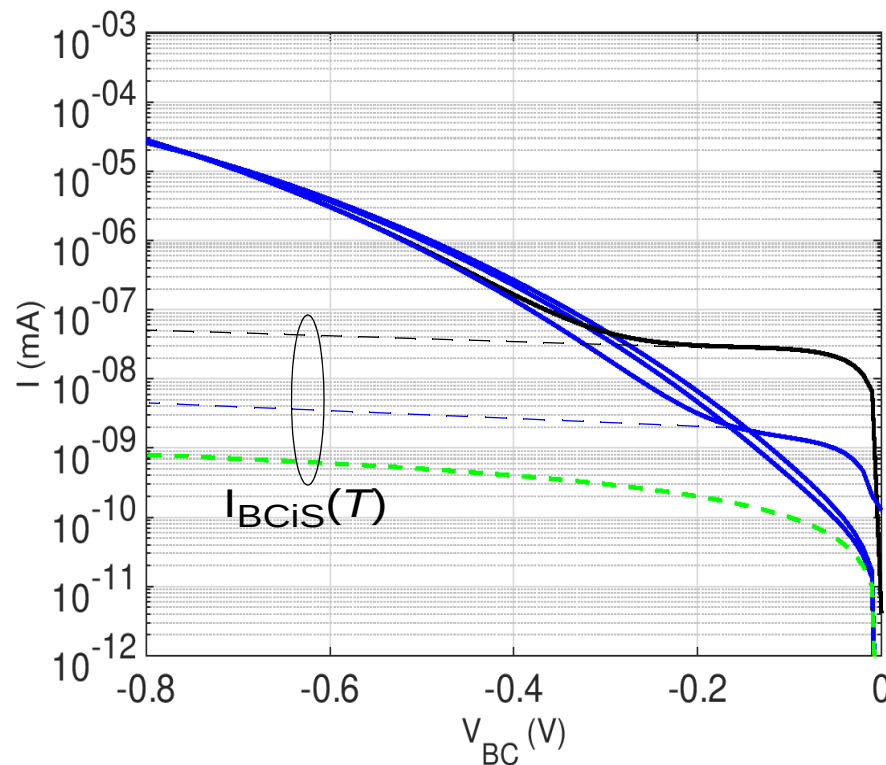
Modified transfer current formulation:

$$i_T = c_{10} \frac{\exp\left(\frac{V_{B'E'}}{V_T}\right) - \exp\left(\frac{V_{B'C'}}{V_T}\right)}{Q_{p0} + h_{jEi}Q_{jEi} + h_{jCi}Q_{jCi} + h_{f0}Q_{f0} + h_{fE}\Delta Q_{Ef} + h_{fB}\Delta Q_{Bf} + h_{fC}\Delta Q_{Cf} + h_{r0}Q_r}$$

- Additional model parameters: **hfb, hr0**
 - default values (for backwards compatibility): $hfb = 1$, $hr0 = 1$
(applies also physically for spatially constant base doping and Ge composition)
 - Temperature dependence unknown yet

Device physics related changes (Q4_2025.1)

Added band-to-band tunneling current to BC branch



- Measurements shows clear evidence of BtB tunneling current through BC junction
- Implemented formulation see documentation
- Test results show correct superposition of BC diode saturation and tunneling current
- Shot noise contribution added

- Adding BtB tunneling current requires two additional model parameters:
 - **ibcts**
 - **abct**

Summary of changes

Coding related changes in v3.2.0 vs. v3.1.0

change	model param.
Defined NIST2018 physical constants in code	- (flcomp)
RTOLC: `define RTOLC \$simparam("reitol",1.0e-5)	-
GMIN: replaced 'GMIN by real variable gmin	-
Base current discontinuity in HICQFF: - eliminated conditional statement (if(itf>=1.0e-6*ick) ...) - re-arranged code in HICQFC to avoid division by zero	-
Re-arranged implementations of saturation current T dependence re. improved computational efficiency	
Declaration of thermal node as <i>thermal</i> (instead of <i>electrical</i>) (& renamed (V,I) -> (Temp,Pwr))	-
Set transfer current iteration tolerance RTOLC=reitol(sim.)	-
Added gmin to SC junction branch dep. on SOI/junction isol.	-

⇒ **Implemented backwards compatible (using flcomp)**

Summary of changes (2)

Device physics related changes in v3.2.0 vs. v3.1.0

change	new model parameters
Add parasitic CE capacitance	ccepar
Addition of BC junction related band-to-band tunneling current	ibcts, abct
Addition of weight factors hfb and hr0 for compensating missing or incorrect values of Qp0 and for improving accuracy in reverse operation	hfb, hr0

⇒ **Implemented backwards compatible
(via default parameter setting)**

- Made implemented intrinsic output conductance for OP output compatible with documentation

Changes in qaSpec

- added test:
 - fgum_npn_1D_dc_vdck
 - reduce max(VC) to 2V:
 - ypara_npn_1D_ac
 - noise_npn_1D_ac
 - ypara_npn_1D_ac_vdck
 - noise_npn_1D_ac_vdck
 - ypara_npn_3D_ac
 - noise_npn_3D_ac
 - ypara_npn_3D_ac_dtf1sh
 - ypara_npn_3D_ac_flcomp
- => reduces no. of base current zero crossings and "false" errors in QA data

Test data

- Reference data were generated with parameters:
ABSTOL=1.0E-14 and GMIN=1e-30
- requires changes at 4 locations in, e.g., file /lib/ads.pm of QA suite
=> should be adjustable in qaSpec

Observations during QA tests

Running QA suite v4.4.0

- model parameter line **+ ccepar = (0.0)** causes error messages

Use of uninitialized value \$value in substitution (s///) at ./lib/modelQaTestRoutines.pm line 515, <IF> line 106.

Use of uninitialized value \$value in substitution (s///) at ./lib/modelQaTestRoutines.pm line 515, <IF> line 106.

Use of uninitialized value \$value in pattern match (m//) at ./lib/modelQaTestRoutines.pm line 516, <IF> line 106.

Use of uninitialized value \$value in pattern match (m//) at ./lib/modelQaTestRoutines.pm line 519, <IF> line 106.

ERROR: model parameter value in ccepar (0.0) is not a number, stopped at ./lib/modelQaTestRoutines.pm line 520, <IF> line 106.

- after changing line to **+ ccepar = (0)**

=> error message disappears

- other lines in parameter file with **+ par = (0.0)** do not cause error message

=> Why is the error message generated?